

Unit 7 – Inference for Quantitative Data: Means

Notes 3: Margin of Error and Matched Pair Procedures

Margin of Error for Means

Often times in studies, we want to determine what sample size we need (before the study is run) in order to ensure a certain margin of error:

$$\text{statistic} \pm \text{margin of error}$$

Our margin of error (ME) for a one sample t interval for means:

Why can we not solve this for n, like we did for a confidence interval for proportions?

- We cannot find t^* without _____, which depends on _____
- We do not know _____, since that relies on _____

Our fixes for these problems:

- Use _____ as an estimate for t^*
- Use a reasonable value for σ based on past studies

Our margin of error to determine a sample size for a confidence interval for a mean:

Example: Determining Sample Size

Medical researchers would like to do a large study to determine the mean cholesterol level μ of children aged 5 to 10. They want their estimate to be within 1 milligram per deciliter (mg/dl) of the true value at a 95% confidence level. Previous studies have shown that the cholesterol level in children of this age varies with a population standard deviation of 5 mg/dl. What is the minimum number of children the researchers need in their sample to get the estimate they want?

Matched Pair Procedures

- Comparative studies are more convincing than single-sample investigations
- For that reason, one-sample inference is less common than _____
- Study designs that involve making two observations on the same individual, or one observation on each of two similar individuals, result in _____

Recall: A _____ is a special case of a randomized block design.

- Using a blocking variable, subjects are arranged in pairs, matched on _____.
- Matched pairs may be formed naturally or by the experimenter.
- Every pair receives _____ by randomly assigning one treatment to one member of the pair and subsequently assigning the remaining treatment to the second member of the pair.
- Alternately, each subject may get both treatments, and the order that the subject gets the treatment is randomly assigned.
- Paired data comes from measuring the _____, on the same subject (individual)
- We can find the difference between the two data points for each subject; we can then analyze the differences
- We use _____ and perform inference on the mean difference:
- These methods are called _____

The inference steps will look very similar to the one sample t procedures we have done:

Communicate	Conditions
$H_0: \mu_d = 0$ (usually 0 but it can be a value!) $H_a: \mu_d \text{ ____ } 0$ $\mu_d =$ in words $\alpha = \text{ ______ }$	<u>Random:</u> Check for random assignment <u>Normal:</u> Given sample data when $n < 30$, graph to check for no outliers or strong skewness (remember you are NOT checking to see that the sample data is “normal”)
Calculate	Conclude
One Sample T Test $t = \frac{\bar{x} - \mu_d}{\frac{s}{\sqrt{n}}}$ p-value → Use tcdf on your calculator in the direction of the alternative, with $df = n - 1$	Because our p-value of _____ is lower/higher than $\alpha = \text{ ______ }$, we reject/fail to reject H_0 . We do/do not have convincing evidence of [the alternative hypothesis in context]

Example: Matched Pair Procedure

Researchers wanted to determine if a new type of medicine will help lower glucose production in individuals with Type 2 diabetes. Because subjects with Type 2 diabetes vary in their eating, exercise, and weight loss journey, researchers decided to do a matched pairs experiment. For one week, individuals will be placed on the new medicine and told to continue with their normal treatment of type 2 diabetes. Their blood sugar will be then be recorded (as a percent). For the next week, individuals will be placed on their current medication for glucose control and told to continue with their normal treatment, and their blood sugar will then be recorded. This experiment will be double blind and the order that the subjects received the medicine will be randomized. The following is a table of their results, along with the difference between current – new:

Current Medicine	6.1	6.0	6.1	6.2	6.2	5.7	5.6	5.5	5.9	5.6	5.8	6.0	6.1	6.3	6.4	6.1
New Medicine	5.9	5.9	6.2	6.0	6.1	5.2	5.3	5.7	6.0	5.3	5.9	5.5	5.4	6.2	6.2	6.3
Difference	0.2	0.1	-0.1	0.2	0.1	0.5	0.3	-0.2	-0.1	0.3	-0.1	0.5	0.7	0.1	0.2	-0.2

Do the data suggest that there is a significant difference between the current medicine and the new medicine for lowering blood sugar in people with type 2 diabetes? Use a 5% level of significance.